



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2023-24/OPN/AE/477944

Date:

12/09/2023(16:17:16)

Subject:LMI for Operating procedure to be followed in case of excess ash accumulation inside ESP hoppers

The LMI for "Operating procedure to be followed in case of excess ash accumulation inside ESP hoppers" Rev-2 revised in line with the latest OGN is being put forward for approval.

The following notes have been incorporated in this latest revision

1. SL.NO:4.2 Rewritten, Added Annexure-1for checklist.

May please be approved.

Submitted by:

सम्राट बनिक - प्रबंधक

SAMRAT BANIK - 103494 (MANAGER ,OPERATION) ,Simhadri
(SAMRATBANIK@NTPC.CO.IN, 9425284908)

On:12/09/2023(16:17:16) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:2

May please be approved.

Submitted by:

सम्राट बनिक - प्रबंधक

SAMRAT BANIK - 103494 (MANAGER ,OPERATION)
(SAMRATBANIK@NTPC.CO.IN, 9425284908) ,Simhadri

On:12/09/2023(16:17:51) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:3

Proposed LMI is reviewed and found in line with OGN.

Submitted by:

मलय कान्ति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:12/09/2023(17:26:13) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:4

The custodian of the LMI may be instructed to present the essence of this LMI and inter departmental responsibilities defined as per the LMI at the earliest in DPM for the wide knowledge of all sectional heads and then may be forwarded for further approval

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)

(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:13/09/2023(10:38:51) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:5

Do needful as per comment from GM(O).

Submitted by:

मलय कान्ति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)

(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:18/09/2023(14:04:06) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:6

Submitted by:

सम्राट बनिक - प्रबंधक

SAMRAT BANIK - 103494 (MANAGER ,MECH MAINT- BOILER)

(SAMRATBANIK@NTPC.CO.IN, 9425284908) ,Simhadri

On:07/11/2023(17:43:52) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:7

presentation done as instructed.

Submitted by:

सतीश कुमार मुरपाक - वरिष्ठ प्रबंधक

SATISH KUMAR MURAPAKA - 101773 (SENIOR MANAGER ,OPERATION)

(SKMURAPAKA@NTPC.CO.IN, 9437479788) ,Simhadri

On:28/11/2023(15:56:47) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:8

Submitted by:

मलय कान्ति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)

(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:28/11/2023(17:25:19) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:9

Submitted by:

ऊमा शंकर वर्मा - अपर महाप्रबंधक

Uma Shanker Verma - 006137 (ADDL. GENERAL MANAGER ,EEMG)
(USVERMA@NTPC.CO.IN, 9650992821) ,Simhadri

On:28/11/2023(17:52:12) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:10

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:30/11/2023(11:44:35) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:11

Please review the LMI

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:30/11/2023(17:12:36) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:12

LMI reviewed, found in order.

Forwarded for further approvals.

Submitted by:

तन्मय सिन्हा - वरिष्ठ प्रबंधक

TANMOY SINHA - 101612 (SENIOR MANAGER ,ASH HANDLING)

(TANMOYSINHA@NTPC.CO.IN, 9434737629) ,Simhadri

On:30/11/2023(17:39:22) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:13

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)

(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:01/12/2023(09:47:14) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:14

Reviewed and forwarded for approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

On:01/12/2023(12:06:06) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:15

Submitted by:

धर्मेंद्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:01/12/2023(16:58:50) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:16

Section 4.1 under IMPLEMENTATION REQUIREMENTS may be reviewed as ALI is now under the scope of C&I as per Unified maintenance system in vogue.

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:05/12/2023(09:36:58) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:17

LMI Docuemnts updated and uploded as per the Note 16.

may please process further.

Submitted by:

सतीश कुमार मुरपाक - वरिष्ठ प्रबंधक

SATISH KUMAR MURAPAKA - 101773 (SENIOR MANAGER ,OPERATION)
(SKMURAPAKA@NTPC.CO.IN, 9437479788) ,Simhadri

On:06/12/2023(13:19:18) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:18

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:11/12/2023(16:50:39) **Ref:**SMTPP/OPERATION/2023-24/OPN/AE/477944

Note No:19

May please be approved

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:11/12/2023(19:35:05) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

Note No:20

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:13/12/2023(11:29:43) **Ref:**SMTTP/OPERATION/2023-24/OPN/AE/477944

NTPC LIMITED

Simhadri Super Thermal Power Station

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/032

Issue No: 1 Rev: 02 Date: 29 August 2023

OPERATING PROCEDURE TO BE FOLLOWED IN CASE OF EXCESS ASH
ACCUMULATION INSIDE ESP HOPPERS

Approved for Implementation by

HOP (Simhadri)

Date:.....

OPN (MAIN PLANT) Department SIMHADRI

LMI	<i>OPERATING PROCEDURE TO BE FOLLOWED IN CASE OF EXCESS ASH ACCUMULATION INSIDE ESP HOPPERS</i>		
LMI No	LMI/OGN/OPS/SYST/032		
Issue No	1	Revision: 2	Dt: 29.08.2023

	Name	Designation	Signature
Prepared by:	MUTHINENI SANJAY	Engineer (O)	
Checked by:	SAMRAT BANIK	Manager(O)	
Checked by:	M.K JANA	AGM (O)	
Rechecked by :	T.R.RAJAGOPALAN	GM (O)	
Checked by	NILANJAN MANDAL	AGM (AHD)	
Checked by	G.HARINARAYANA RAO	AGM (C&I)	
Checked by	DHARMENDRA KUMAR TOMAR	AGM (EMD)	
Rechecked by	MATHEW EIPE KOVOOR	GM (Maint)	
Reviewed by:	DHURJATI PRASAD PATRA	GM (O&M)	
Approved by	SANJAY KUMAR SINHA	HOP (Simhadri)	

CONTENTS

Sl.No.	Description	Page No
1.0	Introduction	4
2.0	Superseded Documents	4
3.0	Scope	4
4.0	Implementation Requirements	4-5
5.0	Records	5
6.0	Responsibilities	5
7.0	Reference document	5
8.0	Review	5
9.0	Annexure-1 Checklist	6-7

OPERATING PROCEDURE TO BE FOLLOWED IN CASE OF EXCESS ASH ACCUMULATION INSIDE ESP HOPPERS

1.0	INTRODUCTION
	In thermal power plants, ash is generated continuously after firing of coal. It is collected in boiler bottom ash hopper, Economizer hoppers, APH, Duct & ESP hoppers. Bottom & ECO hoppers ash is handled by slurry system and APH, Duct & ESP hoppers ash either handled by slurry system or by pneumatic conveying system. In Simhadri Thermal power station, stage1&2 bottom ash hoppers, eco hoppers, stage2 APH hoppers are evacuated/ removed continuously by slurry system. Whereas, Stage1 &2 ESP hoppers and stage1 APH hoppers are evacuated/ removed intermittently by ESP pneumatic conveying system. Each ESP hopper storage capacity of minimum 8 hours ash corresponds to minimum 70% ash collection rate in the first field. Though, in normal condition, 1st field hoppers collect 70-75% of ESP ash, 2nd field hoppers-18%, 3rd field hoppers-5% and so on reduced gradually, for smooth & safe O&M of the system as well as unit, all hoppers emptiness is ensured in every shift. This LMI identifies the need to exercise great care to avoid excess ash accumulation in the ESP hoppers which results in damage of ESP internals & structures. These instructions shall also cover safe operation, isolation of ESP pass/passes and unit shutdown to avoid major damage and long outage.
2.0	SUPERSEDED DOCUMENTS & REVISION NOTES
	LMI/OGN/OPS/SYST/032 Issue no: 1, Rev no: 1, Dt: 01 Jan 2021. Rev Notes: SL.NO:4.2 Rewritten, Added Annexure-1 for checklist.
3.0	SCOPE:
	This LMI is applicable to stage I & II units boilers, with collection of ESP ash in ESP hoppers and removal of same continuously or intermittently either in wet mode or dry mode.
4.0	IMPLEMENTATION REQUIREMENTS:
	Due to hopper chocking and poor ash evacuation rate from ESP hoppers, ash accumulation in the hoppers is taking place. Excess ash accumulation leads to ESP internals damage and collapse of ESP structure. To prevent such incident, below mentioned operating procedure is to be followed:
4.1	ESP hoppers are provided with ash level high indicators (ALI) and all to be in-service before ESP Pass charging. Healthiness of all ALI's to be checked every six months and joint protocol by Operation / C&I/ AHP Maintenance to be made.
	Responsibility: OPN, AHD& EMD and C&I

Operating procedure to be followed in case of excess ash accumulation inside ESP hoppers

4.2	Whenever a) ALI-hi appears in any three fields of a pass or b) If any two fields in series are not getting empty and ALI-Hi persists in a shift, or Under voltage appears in any three fields of a pass and same hoppers not emptied in a shift, or c) Combination of ALI-hi &/or under voltage and not emptied in a shift of any three fields of a pass, It implies that there is poor evacuation of accumulated ash and hence the corresponding ESP pass may be isolated immediately from Electrical & Flue gas side and do required maintenance to ensure safety of ESP internals and structures.
	Responsibility: SCE in consultation with HOD(O&M)
4.3	If the similar ALI high condition/field tripping as in Cl. 4.2 appears in any other pass of the unit same pass also needs to be isolated.
	Responsibility: UCE , SCE
4.4	Load on the unit is to be reduced proportionately after each ESP pass isolation.
	Responsibility: SCE
4.5	The isolated ESP pass to be put back in service only after ensuring the ash evacuation system healthiness, emptying of all hoppers of the pass-through physical inspection of all the fields and restoration of the fields in all respects
	Responsibility: SCE,AHD,EMD
4.6	After isolation of two passes in case of four pass ESP , if the problem of any ESP ash evacuation still persists (ALI is showing Hi for more than 8 hours), i.e., all charged ESP hoppers are not getting evacuated through the system envisaged in every shift, Boiler is to be taken under shutdown for complete evacuation of ESP ash.
	Responsibility: SCE in consultation with HOD(O&M)
5.0	RECORDS
	1) Ash handling engineer Log Book 2) Annexure-1 Checklist
6.0	RESPONSIBILITY
	HOD (O) is responsible for implementation of this LMI
7.0	REFERENCE DOCUMENT
	EOC/OS/2023-24/OGN/294556 Rev. No.: 2 Date: 25/07/2023, Operating Procedure to be followed in Case of Excess Ash Accumulation inside ESP hoppers.
8.0	REVIEW: GM (O) shall arrange for review and take the approval of competent authority. The document will be reviewed 2 yearly basis or as and when required.

Annexure-1 Checklist:

S.No	Description	Status		Frequency	Responsibility	Action plan (if not ok)	Remarks
1	ALI of ESP Hoppers	OK	NOT OK (Mention details in the remark including the duration)	Each shift	AHP-OPN		
2	ESP HOPPER HEATER	OK (If cut in -cut out temp. maintained as per design)	NOT OK (Mention details in the remark)	Fortnightly	AHP-OPN		
3	Healthiness of ESP fields	Ok (Total no. of fields available)	NOT OK (Reason for non-availability of fields, if any)	Daily	AHP-OPN		

*ABOVE CHECKLIST SHOULD BE INCLUDED IN AHP LOGBOOK.